09 · Design your own multimodal task

Agent: Valerian Core – Generic Multimodal Behaviour · **Group:** 3 · **Time:** 60–70 minutes



Focus: Useful mappings from perception to behaviour. **Task control:** Sensor rules and dialogue.

1 · Set up

Open the class Valerian link with your access code. Create a fresh agent above in the heartbeat drawer, select **Cockpit — Valerian**, and connect. Start in **Text**. Agree participation before camera or microphone capture.

2 · Hello, agent

Ask about capabilities. Explore the whole palette:

Inputs: Typed/spoken words; facial-expression estimates; hand signs; social sensing (people, groupings, changes); current weather and forecasts with supplied location context.

Outputs: Words (text/speech), expressive gestures, facial expression, gaze, stillness/energy, and hand poses. Hand input/output uses rock/scissor/paper.

Try one input and two output channels. Inspect **Signals Sensed**, **Diagnostics** → **Activity**, and **Behaviour**. Counts and labels do not establish identity, intent, or feelings; output plans do not prove physical actions.

3 · Design the task

Choose a person, situation, and useful goal. Define success. Select useful inputs and outputs; explore social or environmental context alongside personal cues. You need not use everything.

Sketch **observed condition** → **response channels** → **useful next step** for each mapping. Include a sensor-triggered action. Predict the benefit and identify one cue to ignore or clarify.

4 · Give the instructions

Enable chosen detectors and **Emit camera observations** for camera input. For weather, select a location, **Fetch Current**, then **Send Current** or **Send Forecast**; check place/time. Location is supplied context, not automatic tracking or a rule trigger.

Adapt: “Propose [goal]. Begin with [action]. When [condition], respond with [words and expressive cues]. Otherwise wait. Finish at [completion condition], with at most four actions including the first. Do not start yet.”

Use up to four independent sensor rules, each checking one supported field; use dialogue for other decisions. Review conditions, outputs, and waiting interval; start with one matching sample. Check **Awaiting your acceptance**, then explicitly accept and start.

5 · Test the interaction

Ordinary case: Try each mapping separately. Trace received input, output plan, displayed/audible response, and the person's next action. Did the task progress?

Challenging case: Choose an ambiguous, missing, or irrelevant cue. Predict whether the agent should wait, clarify, or respond. Inspect received evidence before judging behaviour. Use a new accepted run if needed.

6 · Improve and compare

Change one mapping or output instruction. Accept a revised sensor proposal; repeat the challenging episode comparably. Compare success, repairs, effort, and the person's experience. Explain using **shared mental models** or **participatory sensemaking**. Complete the worksheet and say “Stop the task.”

Speech option: After the Text comparison, use the guide's speech notes for a separate spoken trial; record what was audible.

Worksheet · Design, test, improve

Names / class: _____ **Exercise:** _____ **Participant:** _____

Keep the same participant during a comparison; rotate afterwards. Record selected evidence.

1 · Set up

Mode / enabled inputs: _____ **Output actually seen or heard:** _____

2 · Hello, agent

We expected it to perceive or express …; the evidence showed … Separate detected, sent, and used.

_____

3 · Design the task

Person and goal: _____

Success means: _____

We predict this cue or output will help / hinder because: _____

_____

4 · Give the instructions

Our instruction or agreed rule: _____

_____

How we started; limit / stopping condition: _____

5 · Test the interaction

Mark evidence as live, manual input, or simulated. Trace input → response → human action → task progress.

Case	Expected → observed behaviour	Evidence and effect on the task
Ordinary		
Challenging		
After one change		

6 · Improve and compare

One change and its reason: _____

Task result; effort / repairs; experience and usefulness: _____

_____

One HCI explanation; another possible explanation: _____

_____

Next check; another affected person or embodiment: _____

Present in 3 minutes: Task and agreement → evidence → change and comparison → explanation and next check.